

Lab 2: Pipelining and Superscalar using SimpleScalar(2)

October 16, 2025

Introduction

This lab extends the previous lab on pipelining and superscalar architectures. If you have not completed the previous lab, please do so first.

Part 1: Increasing Functional Units (In-Order)

Procedure

1. Make a copy of `default.cfg`, call it `config_c.cfg`. Modify it with the same settings as `config_a.cfg` from the previous lab, except change the resources.

```
cp simplesim-3.0/config/default.cfg labs/config_c.cfg
```

In `labs/config_c.cfg`, ensure the following are set:

```
# From Lab 1, Part 1
-decode:width 1
-fetch:ifqsize 1
-issue:width 1
-issue:inorder true
-ruu:size 8
-lsq:size 8
# New for this part
-res:ialu 4
-res:imult 4
-res:memport 4
-res:fpalu 4
-res:fpmult 4
```

2. Run the simulation with `config_c.cfg`.

```
cd labs
../simplesim-3.0/sim-outorder -config config_c.cfg -redir:sim sim_configc.out
→ ./test-math
```

Questions

1. What is the IPC of the `test-math` program using this configuration?
 2. Compare this configuration and the configuration from Part 1 of the previous lab. Does adding resources increase performance in an in-order pipeline?
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Part 2: Increasing Functional Units (Out-of-Order)

Procedure

1. Make a copy of config_c.cfg, call it config_d.cfg. Change it to be out-of-order.

```
# Make sure you are in the labs directory
cp config_c.cfg config_d.cfg
```

In config_d.cfg, make one change:

```
-issue:inorder false
```

2. Run the simulation with config_d.cfg.

```
../simplesim-3.0/sim-outorder -config config_d.cfg -redir:sim sim_configd.out
→ ./test-math
```

Questions

1. What is the IPC using this configuration?
 2. Compare this configuration with Part 1 of this lab. Does out-of-order execution help when resources are plentiful but pipeline width is narrow?
 3. As the system contains 4 functional units, find the bottleneck. Run multiple simulations based on config_d.cfg, but change only one of these parameters to 4 at a time: -decode:width, -issue:width. Does performance increase?
 4. Based on your findings, what is the main bottleneck in configuration D, and what is your conclusion?
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Part 3: Widening the Pipeline

Procedure

1. Make a copy of default.cfg, call it config_e.cfg. Make the following changes:

```
-fetch:ifqsize 4
-decode:width 4
-lsq:size 8
-ruu:size 8
-issue:width 4
-issue:inorder false
-res:ialu 1
-res:imult 1
-res:memport 1
-res:fpalu 1
-res:fpmult 1
```

2. Run the simulation.

```
../simplesim-3.0/sim-outorder -config config_e.cfg -redir:sim sim_confige.out
→ ./test-math
```

Questions

1. What is the IPC for this configuration?
 2. Compare this configuration with Part 2 of the previous lab (config_b.cfg). What is the effect of widening the pipeline front-end?
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Part 4: Wide Pipeline and Many Resources

Procedure

1. Make a copy of config_e.cfg, call it config_f.cfg. Change the resources.

```
cp config_e.cfg config_f.cfg
```

In config_f.cfg, set all resource types to 4.

```
-res:ialu 4
-res:imult 4
-res:memport 4
-res:fpalu 4
-res:fpmult 4
```

2. Run the simulation.

```
../simplesim-3.0/sim-outorder -config config_f.cfg -redir:sim sim_configf.out
➡ ./test-math
```

Questions

1. What is the IPC for this configuration?
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Part 5: Increasing the Instruction Window

Procedure

1. Make a copy of config_f.cfg, call it config_g.cfg. Change the RUU size.

```
cp config_f.cfg config_g.cfg
```

In config_g.cfg, set:

```
-ruu:size 16
```

2. Run the simulation.

```
../simplesim-3.0/sim-outorder -config config_g.cfg -redir:sim sim_configg.out
➡ ./test-math
```

Questions

1. What is the IPC for this configuration?